

$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#2}$				
$\mathcal{K}_{0^+}^{\#1} \dagger$	$\frac{1}{18} (2 \Upsilon_1 k^2 + 9 \binom{(2)}{K_1})$	$\frac{1}{6} (2 \Upsilon_2 k^2 - 3 \binom{(2)}{K_1})$			
$\mathcal{K}_{0^+}^{\#2} \dagger$	$\frac{1}{6} (2 \Upsilon_2 k^2 - 3 \binom{(2)}{K_1})$	$\frac{1}{18} (2 \Upsilon_1 k^2 + 9 \binom{(2)}{K_1})$			
		$\mathcal{K}_{1^-}^{\#1}{}_\alpha$	$\mathcal{K}_{1^-}^{\#2}{}_\alpha$		
		$\mathcal{K}_{1^-}^{\#1} \dagger^\alpha$	$\frac{1}{18} (-10 \Upsilon_3 k^2 + 15 \binom{(2)}{K_1})$	$\frac{1}{18} (2 \sqrt{5} \Upsilon_3 k^2 - 3 \sqrt{5} \binom{(2)}{K_1})$	
		$\mathcal{K}_{1^-}^{\#2} \dagger^\alpha$	$\frac{1}{18} (2 \sqrt{5} \Upsilon_3 k^2 - 3 \sqrt{5} \binom{(2)}{K_1})$	$\frac{1}{18} (-2 \Upsilon_3 k^2 + 3 \binom{(2)}{K_1})$	
				$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{9} (-2 \Upsilon_4 k^2 + 9 \binom{(2)}{K_1})$
				$\mathcal{K}_{3^-}^{\#1}{}_{\alpha\beta\chi}$	
				$\mathcal{K}_{3^-}^{\#1} \dagger^{\alpha\beta\chi}$	$\frac{1}{9} (-\Upsilon_5 k^2 + 9 \binom{(2)}{K_1})$

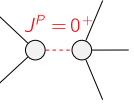
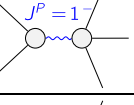
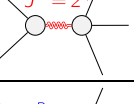
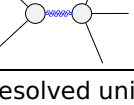
$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#2}$				
$\mathcal{J}_{0^+}^{\#1} \dagger$	$\frac{2 \Upsilon_1 k^2 + 9 \binom{(2)}{K_1}}{18 \text{Det}(0^+)}$	$\frac{-2 \Upsilon_2 k^2 + 3 \binom{(2)}{K_1}}{6 \text{Det}(0^+)}$			
$\mathcal{J}_{0^+}^{\#2} \dagger$	$\frac{-2 \Upsilon_2 k^2 + 3 \binom{(2)}{K_1}}{6 \text{Det}(0^+)}$	$\frac{2 \Upsilon_1 k^2 + 9 \binom{(2)}{K_1}}{18 \text{Det}(0^+)}$	$\mathcal{J}_{1^-}^{\#1}{}_\alpha$	$\mathcal{J}_{1^-}^{\#2}{}_\alpha$	
		$\mathcal{J}_{1^-}^{\#1} \dagger^\alpha$	$\frac{5}{6 \text{Det}(1^-)}$	$\frac{\sqrt{5}}{6 \text{Det}(1^-)}$	
		$\mathcal{J}_{1^-}^{\#2} \dagger^\alpha$	$\frac{\sqrt{5}}{6 \text{Det}(1^-)}$	$\frac{1}{6 \text{Det}(1^-)}$	
				$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	
				$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$
				$\mathcal{J}_{3^-}^{\#1}{}_{\alpha\beta\chi}$	
				$\mathcal{J}_{3^-}^{\#1} \dagger^{\alpha\beta\chi}$	$\frac{1}{\text{Det}(3^-)}$

Abbreviations used in matrices

$$\begin{aligned} \Upsilon_1 &= 9 \binom{(4)}{K_1} - 9 \binom{(4)}{K_2} + 4 \binom{(4)}{K_3} \& \& \Upsilon_2 = 3 \left(\binom{(4)}{K_1} + \binom{(4)}{K_2} \right) - \binom{(4)}{K_3} \& \& \Upsilon_3 = 3 \binom{(4)}{K_2} - \binom{(4)}{K_3} \& \& \Upsilon_4 = 9 \binom{(4)}{K_2} - 2 \binom{(4)}{K_3} \& \& \\ \Upsilon_5 &= 18 \binom{(4)}{K_2} - \binom{(4)}{K_3} \& \& \text{Det}(0^+) = \frac{1}{81} k^2 (18 \binom{(4)}{K_1} + \binom{(4)}{K_3}) (k^2 (-18 \binom{(4)}{K_2} + 7 \binom{(4)}{K_3}) + 9 \binom{(2)}{K_1}) \& \& \\ \text{Det}(1^-) &= \frac{2}{3} k^2 (-3 \binom{(4)}{K_2} + \binom{(4)}{K_3}) + \binom{(2)}{K_1} \& \& \text{Det}(2^+) = k^2 (-2 \binom{(4)}{K_2} + \frac{4 \binom{(4)}{K_3}}{9}) + \binom{(2)}{K_1} \& \& \text{Det}(3^-) = k^2 (-2 \binom{(4)}{K_2} + \frac{4 \binom{(4)}{K_3}}{9}) + \binom{(2)}{K_1} \end{aligned}$$

Lagrangian

$$\begin{aligned} & \binom{(2)}{K_1} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - \frac{1}{2} \binom{(2)}{K_1} \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \mathcal{K}^\chi{}_\chi{}^\chi{}_\chi + \binom{(4)}{K_1} \partial_\beta \mathcal{K}_\chi{}^\delta{}_\delta \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \binom{(4)}{K_2} \partial_\chi \mathcal{K}_\beta{}^\delta{}_\delta \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \\ & \binom{(4)}{K_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_\beta{}^\delta{}_\chi - \frac{2}{3} \binom{(4)}{K_3} \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \partial_\delta \mathcal{K}_\beta{}^\delta{}_\chi - 2 \binom{(4)}{K_2} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \frac{1}{9} \binom{(4)}{K_3} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^-}^{\#2\alpha} + \mathcal{J}_{1^-}^{\#1\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi{}_\chi = \partial_\chi \partial^\chi \mathcal{J}^\alpha{}_\alpha{}^\beta{}_\beta$	
Total # constraint(s):	3		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$\frac{9 \binom{(2)}{K_1}}{18 \binom{(4)}{K_2} - 7 \binom{(4)}{K_3}}$	$-\frac{9}{18 \binom{(4)}{K_2} - 7 \binom{(4)}{K_3}}$
	3	$\frac{3 \binom{(2)}{K_1}}{6 \binom{(4)}{K_2} - 2 \binom{(4)}{K_3}}$	$\frac{3}{6 \binom{(4)}{K_2} - 2 \binom{(4)}{K_3}}$
	5	$\frac{9 \binom{(2)}{K_1}}{18 \binom{(4)}{K_2} - 4 \binom{(4)}{K_3}}$	$-\frac{9}{18 \binom{(4)}{K_2} - 4 \binom{(4)}{K_3}}$
	7	$\frac{9 \binom{(2)}{K_1}}{18 \binom{(4)}{K_2} - \binom{(4)}{K_3}}$	$\frac{9}{18 \binom{(4)}{K_2} - \binom{(4)}{K_3}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	